

COMETS

There are billions of comets at the edge of the solar system, but very few have been observed, since they are bright enough to be seen in Earth's sky only when they travel into the inner solar system and approach the Sun. Nearly 2,000 comets have been recorded and their orbits calculated so far. About 200 of the cataloged comets are periodic, having orbital periods of less than 20 years (short period) or between 20 and 200 years (intermediate period). Most, but not all, comets are named after their discoverers.

COMET HALE-BOPP

Caught in the evening sky above Germany in 1997, Hale-Bopp, one of the brightest comets of the 20th century, clearly has two tails.

INTERMEDIATE-PERIOD COMET

Ikeya-Seki

CLOSEST APPROACH TO THE SUN 290,000 miles (470,000 km)

ORBITAL PERIOD 184 years

FIRST RECORDED September 8, 1965

This comet is named after the two amateur Japanese comet hunters, Ikeya Kaoru and Seki Tsutomu, who discovered it independently (and within 5 minutes of each other) in 1965. On October 21, 1965, as it passed perihelion, the comet was so bright that it was visible in the noon sky only 2 degrees from the Sun. Tidal forces then caused the nucleus to split into three parts. Ikeya-Seki faded quickly as it moved away from the Sun, but the tail grew until it extended over 60 degrees across the sky. At this stage, it was 121 million miles (195 million km) from the Sun.

SUNGRAZER

Ikeya-Seki is a sungrazer and passed within just 290,000 miles (470,000 km) of the Sun's surface in 1965. It is one of over 1,000 comets in the Kreutz sungrazer family.

LONG-PERIOD COMET

Great Comet of 1680

CLOSEST APPROACH TO THE SUN 580,000 miles (940,000 km)

ORBITAL PERIOD 9,400 years

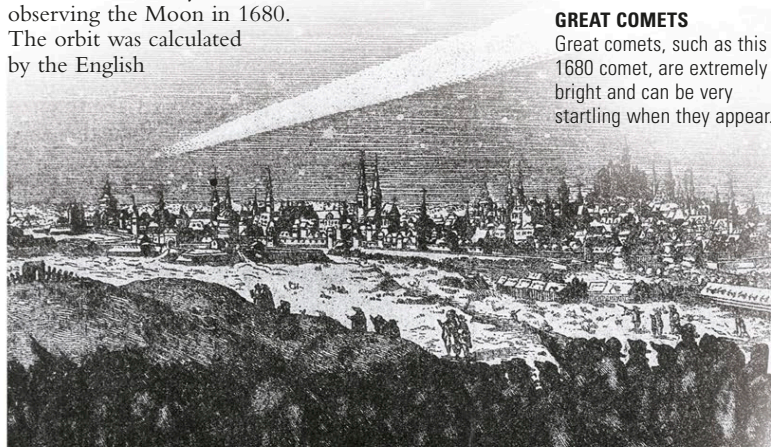
FIRST RECORDED November 14, 1680

This comet has two great claims to fame. It was the first comet to be discovered by telescope and the first to have a known orbit. Some 70 years after the telescope was invented, the German astronomer Gottfried Kirch found the comet by accident when observing the Moon in 1680. The orbit was calculated by the English

mathematician Isaac Newton using his new theory of universal gravity, and the results were published in his masterpiece *Principia* in 1687. The comet is a sungrazer and was seen twice: first, as a morning phenomenon, when it was approaching the Sun; and subsequently in the evenings, when it was receding. Newton was the first to realize that these apparitions were of the same comet. The English physicist Robert Hooke noticed a stream of light issuing from the nucleus. This was the first description of jets of material emanating from active areas.

GREAT COMETS

Great comets, such as this 1680 comet, are extremely bright and can be very startling when they appear.



EXPLORING SPACE

COMET ORBIT

Isaac Newton made observations of the Great Comet of 1680. At the time, a conventional view held that comets traveled in straight lines, passing through the solar system only once. Based on his observations, Newton realized that he had seen a comet traveling around the Sun on a parabolic curve. In 1687, in the *Principia*, he used his study of comets and other phenomena to confirm his law of universal gravitation. He also showed how to calculate a comet's orbit from three accurate observations of its position. Using Newton's laws, Edmond Halley successfully predicted the return of the comet named after him.



NEWTON'S ORBIT SKETCH

INTERMEDIATE-PERIOD COMET

Swift-Tuttle

CLOSEST APPROACH TO THE SUN 88 million miles (143 million km)

ORBITAL PERIOD About 135 years

FIRST RECORDED July 16, 1862

After Swift-Tuttle's discovery in 1862, calculations of its orbit established the relationship between comets and meteoroid streams. Every August, the Earth passes through a stream of dust particles that produces the Perseid meteor shower, named after the constellation from which the shooting stars appear to be emanating.

PERSEIDS

It takes about 2 weeks for the Earth to pass through this meteoroid stream. The peak rate is on August 12, at about 50 visible meteors per hour.



PERIODIC COMET

Swift-Tuttle was discovered independently by American astronomers Lewis Swift and Horace Tuttle in 1862. This optical image was taken in 1992, when the comet approached the Sun once again.

In 1866, Giovanni Schiaparelli (see p.220), the director of the Milan Observatory in Italy, calculated the mean orbit of the Perseid meteoroids. He immediately realized that this orbit was very similar to that of Comet Swift-Tuttle, which intersects Earth's path. He concluded that meteoroid streams were produced by the decay of comets, the meteoroids being no more than cometary dust particles, a fraction of a gram in mass, hitting the Earth's upper atmosphere at velocities of about 134,000 mph (216,000 kph). About the same

number of Perseid meteors are seen each year, so the dust must be evenly spread around the cometary orbit. This uniformity takes a long time to come about. Swift-Tuttle must have passed the Sun on the same orbit a few hundred times to produce this effect. Comets are decaying, but they have to pass through the inner solar system a thousand times or so before they are whittled down to nothing.